

Lista 1

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Zadanie 1

Sprawdzić, że:

$$\begin{aligned} \text{(a)} \quad & \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1 \\ \text{(b)} \quad & \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1-p)^{n-k} = n \sum_{k=1}^n \binom{n-1}{k-1} p p^{k-1} (1-p)^{n-1-(k-1)} \\ & = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np \end{aligned}$$

Zadanie 2

Sprawdzić, że:

$$\begin{aligned} \text{(a)} \quad & \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1 \\ \text{(b)} \quad & \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^k}{(k-1)!} = \lambda \sum_{j=0}^{\infty} e^{-\lambda} \frac{\lambda^j}{j!} = \lambda \end{aligned}$$

Zadanie 3

Funkcją Γ -Eulera nazywamy wartością całki:

$$\Gamma(p) = \int_0^{\infty} t^{p-1} e^{-t} dt, \quad p > 0.$$

Wykazać, że:

$$\Gamma(p) = (p-1)\Gamma(p-1), \quad p \in \mathbb{R}_+$$

$$\begin{aligned} \Gamma(p) &= \int_0^{\infty} t^{p-1} e^{-t} dt \\ \begin{cases} u = t^{p-1}, & dv = e^{-t} dt \\ du = (p-1)t^{p-2} dt, & v = -e^{-t} \end{cases} \end{aligned}$$

$$\Gamma(p) = \left[-t^{p-1} e^{-t} \right]_0^{\infty} + (p-1) \int_0^{\infty} t^{p-2} e^{-t} dt = (p-1)\Gamma(p-1)$$

Podstawa indukcji dla $n = 0$.

$$\Gamma(1) = \int_0^{\infty} t^{1-1} e^{-t} dt = \int_0^{\infty} e^{-t} dt = -e^{-t} \Big|_0^{\infty} = 1 = 0!$$

Zadanie 4

$$(a) \int_0^\infty \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_0^\infty = (0 - (-1)) = 1$$

$$(b) \int_0^\infty x \lambda e^{-\lambda x} dx =$$

$$\begin{cases} u = x, & dv = \lambda e^{-\lambda x} dx \\ du = dx, & v = -e^{-\lambda x} \end{cases}$$

$$= x(-e^{-\lambda x}) \Big|_0^\infty - \int_0^\infty -e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^\infty = \frac{1}{\lambda}$$

Zadanie 5

Wykazać, że $D_n = n$, gdzie

$$D_n = \begin{vmatrix} 1 & -1 & -1 & \cdots & -1 \\ 1 & 1 & & & \\ 1 & & 1 & & \\ \vdots & & & \ddots & \\ 1 & & & & 1 \end{vmatrix} \stackrel{(1)}{=} \begin{vmatrix} n & -1 & -1 & \cdots & -1 \\ 0 & 1 & & & \\ 0 & & 1 & & \\ \vdots & & & \ddots & \\ 0 & & & & 1 \end{vmatrix} \stackrel{(2)}{=} \begin{vmatrix} n & 0 & 0 & \cdots & 0 \\ 0 & 1 & & & \\ 0 & & 1 & & \\ \vdots & & & \ddots & \\ 0 & & & & 1 \end{vmatrix} = n \cdot \det(I_{n-1}) = n$$

$$(1) \quad c_0 := c_0 - c_1 - \dots - c_n, \text{ gdzie } c_i \text{ to } i\text{-ta kolumna } D_n$$

$$(2) \quad r_0 := r_0 + r_1 + \dots + r_n, \text{ gdzie } r_i \text{ to } i\text{-ty wiersz } D_n$$

Zadanie 6

Niech

$$I = \int_{-\infty}^\infty \exp\left(-\frac{x^2}{2}\right) dx.$$

Wówczas

$$I^2 = \left(\int_{-\infty}^\infty \exp\left(-\frac{x^2}{2}\right) dx \right) \left(\int_{-\infty}^\infty \exp\left(-\frac{y^2}{2}\right) dy \right) = \int_{-\infty}^\infty \int_{-\infty}^\infty \exp\left(-\frac{x^2+y^2}{2}\right) dy dx.$$

Stosujemy podstawienie biegunowe

$$x = r \cos \theta, \quad y = r \sin \theta,$$

gdzie $r \in [0, \infty)$, $\theta \in [0, 2\pi)$, a wyznaczone pole wynosi

$$\text{Długość łuku} \times \text{Grubość} = r d\theta \times dr$$

Zatem

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^\infty \exp\left(-\frac{r^2}{2}\right) r dr d\theta \\ &= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^\infty r e^{-r^2/2} dr \right). \end{aligned}$$

W całce po r podstawiamy $u = \frac{r^2}{2}$, wtedy $du = r dr$, więc

$$\int_0^\infty r e^{-r^2/2} dr = \int_0^\infty e^{-u} du = 1.$$

Ostatecznie

$$I^2 = (2\pi) \cdot 1 = 2\pi.$$

Zadanie 7

Udowodnić, że:

$$\begin{aligned} \text{(a)} \quad \sum_{k=1}^n (x_k - \bar{x})^2 &= \sum_{k=1}^n x_k^2 - 2x_k \bar{x} + \bar{x}^2 = \sum_{k=1}^n (x_k^2) - 2n\bar{x}\bar{x} + n\bar{x}^2 = \sum_{k=1}^n x_k^2 - n \cdot \bar{x}^2 \\ \text{(b)} \quad \sum_{k=1}^n (x_k - \bar{x})(y_k - \bar{y}) &= \sum_{k=1}^n x_k y_k - x_k \bar{y} - \bar{x} y_k + \bar{x} \bar{y} = \sum_{k=1}^n (x_k y_k) - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} = \sum_{k=1}^n x_k y_k - n\bar{x}\bar{y} \end{aligned}$$

Zadanie 8

Dane są wektory $\vec{\mu}, X \in \mathbb{R}^n$ oraz macierz $\Sigma \in \mathbb{R}^{n \times n}$. Niech $S = (X - \vec{\mu})^T \Sigma^{-1} (X - \vec{\mu})$ oraz $Y = A \cdot X$, gdzie macierz A jest odwracalna. Sprawdzić, że $S = (Y - A\vec{\mu})^T (A\Sigma A^T)^{-1} (Y - A\vec{\mu})$.

$$\begin{aligned} S &= (Y - A\vec{\mu})^T (A\Sigma A^T)^{-1} (Y - A\vec{\mu}) \\ &= Y^T (A\Sigma A^T)^{-1} Y - (A\vec{\mu})^T (A\Sigma A^T)^{-1} Y \\ &\quad - Y^T (A\Sigma A^T)^{-1} A\vec{\mu} + (A\vec{\mu})^T (A\Sigma A^T)^{-1} A\vec{\mu} \\ &= X^T \Sigma^{-1} X - \vec{\mu}^T \Sigma^{-1} X - X^T \Sigma^{-1} \vec{\mu} + \vec{\mu}^T \Sigma^{-1} \vec{\mu} \\ &= (X - \vec{\mu})^T \Sigma^{-1} (X - \vec{\mu}). \end{aligned}$$

Zadanie 9

Udowodnić, że:

$$\int_0^\infty x^k \lambda \exp(-\lambda x) dx = \frac{k!}{\lambda^k}, \quad k = 0, 1, \dots, \quad \lambda > 0$$

Podstawa indukcji $k = 0, 1$: (zadanie 4).

Indukcja $k \implies k + 1$:

$$\begin{aligned} &\int_0^\infty x^k \lambda \exp(-\lambda x) dx = \\ &\quad \begin{cases} u = x^k, & dv = \lambda e^{-\lambda x} dx \\ du = kx^{k-1} dx, & v = -e^{-\lambda x} \end{cases} \\ &= x^k (-e^{-\lambda x}) \Big|_0^\infty - \int_0^\infty kx^{k-1} (-e^{-\lambda x}) dx = \frac{k}{\lambda} \int_0^\infty x^{k-1} \lambda e^{-\lambda x} dx = \frac{k}{\lambda} \frac{(k-1)!}{\lambda(k-1)} = \frac{k!}{\lambda^k} \end{aligned}$$

Zadanie 10

Obliczyć wartości całki

$$\begin{aligned} \int_0^\infty x \exp\left(-\frac{x^2}{2}\right) dx &= \\ \left\{ u = \frac{x^2}{2}, \quad du = x dx \right. & \\ \int_0^\infty e^{-u} du = -e^{-u} \Big|_0^\infty = 0 - (-1) = 1 \end{aligned}$$